Lecture Notes for MATH 1024
Honors Calculus II

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Last updated: March 9, 2026

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※ Jordan Measure

Definition 1.1 (Jordan measure on rectangles)**.** Let $< a, b > \times < c, d > \subset \mathbb{R}^2$ be a rectangle. We define the *Jordan measure* μ of this rectangle to be

$$\mu(< a, b > \times < c, d >) := (b - a)(d - c).$$

where we interpret $< a, b >$ as one of the following intervals $[a, b], (a, b), [a, b), (a, b]$.

Definition 1.2 (Simple sets)**.** A set $S \subseteq \mathbb{R}^2$ is called a *simple set* if it can be expressed as a finite union of rectangles.

$$S = \bigcup_{i=1}^{n} R_i \quad \text{where each } R_i \text{ is a rectangle.}$$

We define the Jordan measure μ of a simple set S to be

$$\mu(S) := \sum_{i=1}^{n} \mu(R_i).$$

Note that this definition depends on the choice of rectangles R_i that we use to express S as a finite intersection of rectangles. We will show that this definition is well-defined, i.e. the value of $\mu(S)$ does not depend on the choice of rectangles R_i.

Proposition 1.3. The Jordan measure μ of a simple set S is well-defined.

Proof. Let S be a simple set. Suppose we have two different ways of expressing S as a finite union of rectangles:

$$S = \bigcup_{i=1}^{n} R_i = \bigcup_{j=1}^{m} R'_j.$$

We want to show that $\sum_{i=1}^{n} \mu(R_i) = \sum_{j=1}^{m} \mu(R'_j)$.

Consider the collection of rectangles $\{R_i\}_{i=1}^{n}$ and $\{R'_j\}_{j=1}^{m}$. We can form the collection of all possible intersections of these rectangles:

$$\{R_i \cap R'_j : 1 \leq i \leq n, 1 \leq j \leq m\}.$$

Each intersection $R_i \cap R'_j$ is either empty or a rectangle (or a finite union of rectangles). Moreover, we have

$$S = \bigcup_{i=1}^{n} R_i = \bigcup_{j=1}^{m} R'_j = \bigcup_{i=1}^{n} \bigcup_{j=1}^{m} (R_i \cap R'_j).$$

Since the Jordan measure is additive over disjoint unions of rectangles, we can write

$$\sum_{i=1}^{n} \mu(R_i) = \sum_{i=1}^{n} \sum_{j=1}^{m} \mu(R_i \cap R'_j) = \sum_{j=1}^{m} \sum_{i=1}^{n} \mu(R_i \cap R'_j) = \sum_{j=1}^{m} \mu(R'_j).$$

Therefore, $\mu(S)$ is well-defined and does not depend on the choice of rectangles R_i. □

Definition 1.4. Let $X \subset \mathbb{R}^2$ be a bounded set. We define the inner μ_* and outer μ^* Jordan measures of X as follows:

$$\mu_*(X) := \sup\{\mu(A) : \underset{\text{simple}}{A} \subset X\}$$

$$\mu^*(X) := \inf\{\mu(B) : \underset{\text{simple}}{B} \supset X\}$$

Remark 1.5. Note that sup and inf exist since the set of simple sets contained in X is non-empty (e.g. the empty set) and the set of simple sets containing X is also non-empty (e.g. a large rectangle containing X).

Remark 1.6. The definitions of $\mu_*(X)$ and $\mu^*(X)$ are equivalent to these definitions:

$$\forall \varepsilon, \exists \underset{\text{simple}}{A} \text{ such that } \mu_*(X) - \varepsilon < \mu(A) \leq \mu_*(X)$$
$$\forall \varepsilon, \exists \underset{\text{simple}}{B} \text{ such that } \mu^*(X) + \varepsilon > \mu(B) > \mu^*(X)$$

Remark 1.7. For all $X \subset \mathbb{R}^2$, we have $\mu_*(X) \leq \mu^*(X)$.

Proof. For all $\varepsilon > 0$, there exist simple sets A and B with $A \subset X \subset B$ such that

$$\mu_*(X) - \varepsilon < \mu(A) \leq \mu(B) < \mu^*(X) + \varepsilon$$

(The order on $\mu(A)$ and $\mu(B)$ is trivial). Since ε is arbitrary, we have $\mu_*(X) \leq \mu^*(X)$. □

Definition 1.8. A set $X \subset \mathbb{R}^2$ is called *Jordan measurable* if $\mu_*(X) = \mu^*(X)$. In this case, we define the Jordan measure $\mu(X)$ of X to be $\mu_*(X) = \mu^*(X)$. Otherwise, we say that X is *not Jordan measurable*.

Theorem 1.9. X is Jordan measurable if and only if for all $\varepsilon > 0$, there exist simple sets A and B such that $A \subset X \subset B$ and $\mu(B) - \mu(A) < \varepsilon$.

Proof. ($\implies$) Suppose X is Jordan measurable. Then for all $\varepsilon > 0$, there exist simple sets A and B such that $A \subset X \subset B$ and

$$\mu(X) - \varepsilon/2 < \mu(A) \leq \mu(X) \leq \mu(B) < \mu(X) + \varepsilon/2$$

Rearranging, we get

$$0 \leq \mu(B) - \mu(A) < \varepsilon$$

($\impliedby$) Now, suppose for all $\varepsilon > 0$, there exist simple sets A and B such that $A \subset X \subset B$ and $\mu(B) - \mu(A) < \varepsilon$. Then

$$\mu(A) \leq \mu_*(X) \leq \mu^*(X) \leq \mu(B) \implies 0 \leq \mu^*(X) - \mu_*(X) \leq \mu(B) - \mu(A) < \varepsilon$$

Since ε is arbitrary, we get $\mu_*(X) = \mu^*(X)$. □

Example 1.10. Simple sets are Jordan measurable (take $A = X = B$).

Example 1.11. A point and finitely many points have measure zero.

Theorem 1.12. X is Jordan measurable if and only if there exist simple sets A_n and B_n such that $A_n \subset X \subset B_n$ and $\lim_{n\to\infty} \mu(B_n) = \lim_{n\to\infty} \mu(A_n)$.

Proof. ($\Longrightarrow$) Suppose X is Jordan measurable. For all n, choose $\varepsilon = \frac{1}{n}$ and take simple sets A_n and B_n such that $A_n \subset X \subset B_n$ and $\mu(B_n) - \mu(A_n) < 1/n$. Then

$$0 \leq \mu(B_n) - \mu(A_n) < 1/n \implies \lim_{n\to\infty} \mu(B_n) = \lim_{n\to\infty} \mu(A_n).$$

($\Longleftarrow$) Now, suppose there exist simple sets A_n and B_n such that $A_n \subset X \subset B_n$ and $\lim_{n\to\infty} \mu(B_n) = \lim_{n\to\infty} \mu(A_n)$. Then for all $\varepsilon > 0$, there exists N such that for all $n \geq N$, we have

$$|\mu(B_n) - \mu(A_n)| < \varepsilon \implies \mu(B_n) - \mu(A_n) < \varepsilon$$

choose any $n > N$ with $A = A_n$ and $B = B_n$ to get $A \subset X \subset B$ and $\mu(B) - \mu(A) < \varepsilon$. □

Example 1.13. Let $X = (\mathbb{Q} \cap [0,1])^2 = \{(x,y) \in \mathbb{Q}^2 \mid 0 \leq x, y \leq 1\}$. Then X is not Jordan measurable.

Proof. We first consider the inner measure $\mu_*(X)$. Take any rectangle $R =< a, b > \times < c, d >$ such that $R \subset X$. Since irrationals are dense in $\mathbb{R}$, there exists an irrational number x_0 such that $a < x_0 < b$. Then for all $y \in \mathbb{R}$, we have $(x_0, y) \notin X$. Therefore, $a = b$ and R can at most be a line segment with measure zero. Since this holds for all rectangles $R \subset X$, we have $\mu_*(X) = 0$. Now, consider the outer measure $\mu^*(X)$. Take any rectangle $R =< a, b > \times < c, d >$ such that $X \subset R$. Then R must contain the square $[0,1]^2$ since X is dense in $[0,1]^2$. Therefore, $\mu^*(X) \geq \mu([0,1]^2) = 1$. Since $\mu_*(X) = 0 < 1 \leq \mu^*(X)$, we have $\mu_*(X) \neq \mu^*(X)$ and X is not Jordan measurable. □

Proposition 1.14. For any bounded $X \subset Y \subset \mathbb{R}^2$, we have $\mu_*(X) \leq \mu_*(Y)$ and $\mu^*(X) \leq \mu^*(Y)$. If X, Y are Jordan measurable, then $\mu(X) \leq \mu(Y)$.

Proof.

$$\begin{aligned}\{\mu(A) : A \subset X\} \subseteq \{\mu(A) : A \subset Y\} &\implies \sup\{\mu(A) : A \subset X\} \leq \sup\{\mu(A) : A \subset Y\} \\ &\iff \mu_*(X) \leq \mu_*(Y)\end{aligned}$$

$$\begin{aligned}\{\mu(B) : B \supset X\} \subseteq \{\mu(B) : B \supset Y\} &\implies \sup\{\mu(B) : B \supset X\} \leq \sup\{\mu(B) : B \supset Y\} \\ &\iff \mu^*(X) \leq \mu^*(Y)\end{aligned}$$

□

Theorem 1.15 (Finite subadditivity of the outer measure)**.** Let $X, Y \in \mathbb{R}^2$ be bounded. Then $\mu^*(X \cup Y) \leq \mu^*(X) + \mu^*(Y)$.

Proof. For all $\varepsilon > 0$, there are simple sets $A \supset X$ and $A' \supset Y$ such that $\mu(A) \leq \mu^*(X) + \varepsilon/2$ and $\mu(A') \leq \mu^*(Y) + \varepsilon/2$. Then

$$\mu^*(X) + \mu^*(Y) \geq \mu(A) + \mu(A') - \varepsilon \geq \mu(A \cup A') - \varepsilon \geq \mu^*(X \cup Y) - \varepsilon$$

Since ε is arbitrary, we have $\mu^*(X \cup Y) \leq \mu^*(X) + \mu^*(Y)$. □

Theorem 1.16. If X and Y are Jordan measurable, then

- $X \cup Y$ is Jordan measurable
- $X \cap Y$ is Jordan measurable
- $\mu(X \cup Y) = \mu(X) + \mu(Y) - \mu(X \cap Y)$

Proof. ($X \cup Y$ is measurable) For all $\varepsilon > 0$, there exist simple sets $A \subset X \subset B$ and $A' \subset Y \subset B'$ such that

$$\mu(B) - \mu(A) < \varepsilon/2 \quad \mu(B') - \mu(A') < \varepsilon/2$$

and

$$A \cup A' \subset X \cup Y \subset B \cup B' \implies \mu_*(A \cup A') \leq \mu_*(X \cup Y) \leq \mu^*(X \cup Y) \leq \mu^*(B \cup B')$$

Since $A \cup A'$ and $B \cup B'$ are simple and $A \cup A' \subset B \cup B'$, we have

$$\begin{aligned}
\mu(B \cup B') - \mu_*(A \cup A') &= \mu(B \cup B' \setminus A \cup A') \\
&= \mu((B \setminus A) \cup (B' \setminus A')) \\
&\leq \mu(B \setminus A) + \mu(B' \setminus A') \\
&= \mu(B) - \mu(A) + \mu(B') - \mu(A') \\
&< \varepsilon/2 + \varepsilon/2 \\
&= \varepsilon
\end{aligned}$$

Therefore, $\mu^*(X \cup Y) - \mu_*(X \cup Y) < \varepsilon$.
($X \cap Y$ is measurable) Let $A_n \subset X$ and $A'_n \subset Y$ be simple sets such that

$$\lim_{n\to\infty} \mu(A_n) = \mu(X) \quad \text{and} \quad \lim_{n\to\infty} \mu(A'_n) = \mu(Y)$$

Using the finite subadditivity for simple sets, we have

$$\mu(A_n \cup A'_n) = \mu(A_n) + \mu(A'_n) - \mu(A_n \cap A'_n)$$

Since $\lim_{n\to\infty} \mu(A_n \cup A'_n) = \mu(X \cup Y)$, the limit of $\mu(A_n \cap A'_n)$ must also exist. Similarly for $B_n \supset X$ and $B'_n \supset Y$. The limit argument shows the finite additivity. □

※ Darboux & Riemann Integration

Definition 2.1. A function $f : \mathbb{R} \to \mathbb{R}$ is Darboux integrable on $[a, b]$ if

$$G_{[a,b]}(f) = \left\{(x, y) \mid a \leq x \leq b, 0 \leq y \leq f(x) \text{ or } f(x) \leq y \leq 0\right\}$$

is Jordan measurable. In this case, we define the Darboux integral of f on $[a, b]$ as

$$\int_a^b f(x)\,\mathrm{d}x := \mu(G_{[a,b]}(f) \cap H_+) - \mu(G_{[a,b]}(f) \cap H_-).$$

where $H_+ = \{(x, y) \in \mathbb{R}^2 \mid y \geq 0\}$ and $H_- = \{(x, y) \in \mathbb{R}^2 \mid y \leq 0\}$ are the upper and lower half-planes, respectively.

Definition 2.2 (Partition of an interval)**.** A partition of the interval $[a, b]$ is a finite set of points $P = \{x_0, x_1, \ldots, x_n\}$ such that $a = x_0 < x_1 < \ldots < x_n = b$. The subintervals of P are the intervals $[x_{i-1}, x_i]$ for $i = 1, \ldots, n$. The mesh of the partition P is defined as $\|P\| = \max\limits_{1 \leq i \leq n}(x_i - x_{i-1})$.

Definition 2.3 (Lower and upper Darboux sums)**.** Let $f : [a, b] \to \mathbb{R}$ be a bounded function and $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$. The lower Darboux sum of f with respect to the partition P is defined as

$$L(P, f) = \sum_{i=1}^{n} m_i(x_i - x_{i-1}),$$

where $m_i = \inf\limits_{x \in [x_{i-1}, x_i]} f(x)$ is the infimum of f on the subinterval $[x_{i-1}, x_i]$. The upper Darboux sum of f with respect to the partition P is defined as

$$U(P, f) = \sum_{i=1}^{n} M_i(x_i - x_{i-1}),$$

where $M_i = \sup\limits_{x \in [x_{i-1}, x_i]} f(x)$ is the supremum of f on the subinterval $[x_{i-1}, x_i]$.

Theorem 2.4 (Riemann integrability criterion)**.** A bounded function $f : [a, b] \to \mathbb{R}$ is Darboux integrable on $[a, b]$ if and only if for every $\varepsilon > 0$, there exists a partition P of $[a, b]$ such that

$$U(P, f) - L(P, f) < \varepsilon.$$

Definition 2.5 (Lower and Upper Darboux integrals)**.** Let $f : [a, b] \to \mathbb{R}$ be a bounded function. The lower Darboux integral of f on $[a, b]$ is defined as

$$\underline{\int_a^b} f(x)\,\mathrm{d}x = \sup_P L(P, f),$$

where the supremum is taken over all partitions P of $[a, b]$. The upper Darboux integral of f on $[a, b]$ is defined as

$$\overline{\int_a^b} f(x)\,\mathrm{d}x = \inf_P U(P, f),$$

where the infimum is taken over all partitions P of $[a, b]$.

Definition 2.6 (Uniform continuity)**.** A function $f : [a, b] \to \mathbb{R}$ is uniformly continuous on $[a, b]$ if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $x, y \in [a, b]$, if $|x - y| < \delta$, then $|f(x) - f(y)| < \varepsilon$.

Theorem 2.7. Let f be a bounded function on $[a, b]$. If f is continuous on $[a, b]$, then f is uniformly continuous on $[a, b]$.

Proof. Assume that f is not uniformly continuous. Then there exists $\varepsilon > 0$ such that for all $\delta = \frac{1}{n} > 0$, there exist $x_n, y_n \in [a, b]$ such that

$$|x_n - y_n| < \frac{1}{n} \implies |f(x_n) - f(y_n)| \geq \varepsilon$$

Since x_n and y_n are bounded, by Bolzano-Weierstrass Theorem, there exists x_{n_k} with $\lim_{k\to\infty} x_{n_k} = L \in [a, b]$. Then we have $|x_{n_k} - y_{n_k}| < \frac{1}{n_k}$, so $\lim_{k\to\infty} y_{n_k} = L$. Since f is continuous at L, we have

$$\lim_{k\to\infty} f(x_{n_k}) = f(L) = \lim_{k\to\infty} f(y_{n_k})$$

which contradicts $|f(x_{n_k}) - f(y_{n_k})| \geq \varepsilon$ for all k. This contradiction shows that f must be uniformly continuous. □

Theorem 2.8. If $f : [a, b] \to \mathbb{R}$ is uniformly continuous, then f is Darboux integrable on $[a, b]$.

Proof. For all $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in [a, b]$, if $|x - y| < \delta$, then $|f(x) - f(y)| < \varepsilon$. Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$ such that $\|P\| < \delta$. Then for all $i = 1, \ldots, n$, we have $M_i - m_i < \varepsilon$. Therefore,

$$U(P, f) - L(P, f) = \sum_{i=1}^{n} (M_i - m_i)(x_i - x_{i-1}) < \varepsilon \sum_{i=1}^{n} (x_i - x_{i-1}) = \varepsilon(b - a).$$

Substituting ε with $\varepsilon/(b - a)$, we get $U(P, f) - L(P, f) < \varepsilon$. By the Riemann integrability criterion, f is Darboux integrable on $[a, b]$. □

Definition 2.9 (Riemann sum)**.** Let P^* be a tagged partition of $[a,b]$, i.e., $P^* = \{(x_0, x_1, \ldots, x_n), (t_1, t_2, \ldots, t_n)\}$ where $P = \{x_0, x_1, \ldots, x_n\}$ is a partition of $[a,b]$ and $x_i^* \in [x_{i-1}, x_i]$ is a tag for each subinterval. The Riemann sum of f with respect to the tagged partition P^* is defined as

$$S(P^*, f) = \sum_{i=1}^{n} f(x_i^*)(x_i - x_{i-1}).$$

Remark 2.10. $L(P,f) \leq S(P^*, f) \leq U(P,f)$ for any tagged partition P^* of P.

Definition 2.11. A bounded function $f : [a,b] \to \mathbb{R}$ is Riemann integrable on $[a,b]$ if there exists a number $s \in \mathbb{R}$ such that for every $\varepsilon > 0$, there exists a $\delta > 0$ such that for all tagged partitions P^* of $[a,b]$ with $\|P\| < \delta$, we have

$$|S(P^*, f) - s| < \varepsilon.$$

Theorem 2.12. A bounded function $f : [a,b] \to \mathbb{R}$ is Darboux integrable on $[a,b]$ if and only if f is Riemann integrable on $[a,b]$.

Proof. We will prove that the following statements are equivalent:

1. $\forall \varepsilon > 0, \exists P$ such that $U(P,f) - L(P,f) < \varepsilon$. (Darboux integrability)
2. $\forall \varepsilon > 0, \exists \delta > 0$ such that for all partitions P with $\|P\| < \delta$, we have $|U(P,f) - s| < \varepsilon$ and $|L(P,f) - s| < \varepsilon$ for some $s \in \mathbb{R}$.
3. $\forall \varepsilon > 0, \exists \delta > 0$ such that for all tagged partitions P^* with $\|P\| < \delta$, we have $|S(P^*, f) - s| < \varepsilon$ for some $s \in \mathbb{R}$. (Riemann integrability)

- 2 $\implies$ 3: For all $\varepsilon > 0$, we have

$$s - \varepsilon < L(P,f) \leq S(P^*, f) \leq U(P,f) < s + \varepsilon \implies |S(P^*, f) - s| < \varepsilon.$$

- 2 $\implies$ 1: Choose any partition P with $\|P\| < \delta$. Then we have

$$U(P,f) < s + \varepsilon/2 \text{ and } L(P,f) > s - \varepsilon/2 \implies U(P,f) - L(P,f) < \varepsilon.$$

- 3 $\implies$ 2: For all $\varepsilon > 0$, there exists $\delta > 0$ such that for all tagged partitions P^* with $\|P\| < \delta$, we have $|S(P^*, f) - s| < \varepsilon$. We want to construct a tagged partition P^* such that $\|P\| < \delta$ and $|U(P,f) - s| < \varepsilon$ and $|L(P,f) - s| < \varepsilon$. On $[x_{i-1}, x_i]$ we pick x_i^* such that

$$\sup_{x \in [x_{i-1}, x_i]} f(x) - \varepsilon < f(x_i^*)$$

Then

$$U(P,f) - S(P^*,f) = \sum_{i=1}^{n}(M_i - f(x_i^*))(x_i - x_{i-1}) < \varepsilon \sum_{i=1}^{n}(x_i - x_{i-1}) = \varepsilon(b-a).$$

Therefore,

$$|U(P,f) - s| \leq |U(P,f) - S(P^*,f)| + |S(P^*,f) - s| < \varepsilon(b-a) + \varepsilon = \varepsilon(b-a+1).$$

We can similarly show that $|L(P,f) - s| < \varepsilon(b-a+1)$. Substituting ε with $\varepsilon/(b-a+1)$, we get $|U(P,f) - s| < \varepsilon$ and $|L(P,f) - s| < \varepsilon$.

- 1 $\implies$ 2: Let us first consider the upper sum. Let $s = \inf_P U(P,f)$. By assumption, for all $\varepsilon > 0$, there exists a partition $P_0 = \{a = x_0 < x_1 < \cdots < x_n = b\}$ such that $|U(P_0,f) - s| < \varepsilon$.
  We want to show that there exists $\delta > 0$ such that for all partitions P' with $\|P'\| < \delta$, we have $U(P',f) < s + \varepsilon$. Consider any partition P with $\|P\| < \delta$. There are at most $2(n-1)$ subintervals I_k of P containing some x_i. Let $\mathcal{I}$ be the collection of these subintervals.
  Let $R = P \cup P_0$ be the common refinement of P and P_0. We have

$$s \leq U(R,f) \leq U(P,f) \quad \text{and} \quad s \leq U(R,f) \leq U(P_0,f)$$

  Then

$$\begin{aligned} U(P,f) - U(R,f) &\leq \sum_{I_k \in \mathcal{I}} \mu(\text{rectangles over } I_k) \\ &< 2n \times 2M\delta \end{aligned}$$

  where M is an upper bound for f. Choosing $\delta < \varepsilon/(4nM)$, we get $U(P,f) - U(R,f) < \varepsilon$. Finally,

$$\begin{aligned} |U(P,f) - s| &\leq |U(P,f) - U(R,f)| + |U(R,f) - s| \\ &\leq |U(P,f) - U(R,f)| + |U(P_0,f) - s| \\ &< \varepsilon + \varepsilon = 2\varepsilon. \end{aligned}$$

  Similar arguments show that there exists $\delta > 0$ such that for all partitions P' with $\|P'\| < \delta$, we have $|L(P',f) - s'| < \varepsilon$ where $s' = \sup_P L(P,f)$. Since $s' \leq s$, we have $|L(P',f) - s| < \varepsilon$ as well. Substituting ε with $\varepsilon/2$, we get the desired result.

$\square$

Theorem 2.13. Let f, g be integrable on $[a, b]$. Then $f + g$ is integrable on $[a, b]$ and

$$\int_a^b (f(x) + g(x))\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x + \int_a^b g(x)\,\mathrm{d}x.$$

$$\int_a^b \lambda f(x)\,\mathrm{d}x = \lambda \int_a^b f(x)\,\mathrm{d}x \quad \text{for all } \lambda \in \mathbb{R}.$$

Proof. We have

$$\begin{aligned} S(P^*, f + g) &= \sum_{i=1}^{n} (f(x_i^*) + g(x_i^*))(x_i - x_{i-1}) \\ &= \sum_{i=1}^{n} f(x_i^*)(x_i - x_{i-1}) + \sum_{i=1}^{n} g(x_i^*)(x_i - x_{i-1}) \\ &= S(P^*, f) + S(P^*, g). \end{aligned}$$

Since

$$\lim_{\|P\| \to 0} S(P^*, f) = \int_a^b f(x)\,\mathrm{d}x \quad \text{and} \quad \lim_{\|P\| \to 0} S(P^*, g) = \int_a^b g(x)\,\mathrm{d}x,$$

the limit exists on the left-hand side. Thus we take the limit of both sides and obtain

$$\int_a^b (f(x) + g(x))\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x + \int_a^b g(x)\,\mathrm{d}x.$$

Now, let $\lambda \in \mathbb{R}$. Then

$$\begin{aligned} S(P^*, \lambda f) &= \sum_{i=1}^{n} \lambda f(x_i^*)(x_i - x_{i-1}) \\ &= \lambda \sum_{i=1}^{n} f(x_i^*)(x_i - x_{i-1}) \\ &= \lambda S(P^*, f). \end{aligned}$$

We take the limit as $\|P\| \to 0$ on both sides and get

$$\int_a^b \lambda f(x)\,\mathrm{d}x = \lambda \int_a^b f(x)\,\mathrm{d}x.$$

□

Theorem 2.14. Let f be bounded on $[a, b]$. If f has finitely many discontinuities on $[a, b]$, then f is integrable on $[a, b]$.

Proof. Let S be the set of discontinuities of f. Since $|S|$ is finite, suppose $|S| = N$. For all $\varepsilon > 0$, define

$$I_s = (s - \varepsilon/2, s + \varepsilon/2), \quad s \in S.$$

Then consider the area in $\cup_{s \in S} I_s$. Since $|f| \leq M$ for some $M \in \mathbb{R}$, the area is bounded by $\varepsilon \cdot 2MN$. Since the remaining $N + 1$ intervals are uniformly continuous, the partition can be chosen fine enough to make $U(P, f) - L(P, f) < \varepsilon$ for each interval. Then for the whole interval, we have

$$U(P, f) - L(P, f) < (N + 1)\varepsilon + (2MN)\varepsilon = (2MN + N + 1)\varepsilon.$$

Substituting $\varepsilon \mapsto \varepsilon/(2MN + N + 1)$ gives the desired result. □

Remark 2.15. If f has infinitely many discontinuities on $[a, b]$, this doesn't necessarily imply that f is not integrable.

Remark 2.16. If f is integrable on $[a, b]$, then f is bounded on $[a, b]$.

Proof. Assume that f is unbounded. Then for all $M > 0$, there exists $x_M \in [a, b]$ such that $|f(x_M)| > M$. Let $\varepsilon = 1$. For all $\delta > 0$, there exists a tagged partition $P^* = \{(x_0, x_1, ..., x_n)\}$ such that $\|P\| < \delta$ and $|S(P^*, f) - s| < \varepsilon$. Since $\|P\| < \delta$, there exists i such that $x_M \in [x_{i-1}, x_i]$. We can choose the tag x_i^* to be x_M. Then

$$|S(P^*, f)| \geq |f(t_i)(x_i - x_{i-1})| > M(x_i - x_{i-1}) > 0.$$

By choosing M large enough, we can make $|S(P^*, f)|$ arbitrarily large, which contradicts the existence of the limit of Riemann sums. Therefore, f must be bounded. □

Theorem 2.17. If f is integrable on $[a, b]$, then f is integrable on $[c, d]$ for all $[c, d] \subset [a, b]$.

Proof. For all $\varepsilon > 0$, there exists a P such that $U(P, f) - L(P, f) < \varepsilon$ on $[a, b]$. Let $P_1 = P \cup \{c, d\}$ be the common refinement of P and $\{c, d\}$ and $P_2 = P_1 \cap [c, d]$. Then we have

$$U(P_2, f) - L(P_2, f) < U(P_1, f) - L(P_1, f) < U(P, f) - L(P, f) < \varepsilon.$$

□

Theorem 2.18. Let f be integrable on $[a, b]$ and $c \in [a, b]$. Then f is integrable on $[a, c]$ and $[c, b]$, and

$$\int_a^b f(x)\,\mathrm{d}x = \int_a^c f(x)\,\mathrm{d}x + \int_c^b f(x)\,\mathrm{d}x.$$

Proof. For all partitions P of $[a,b]$, let $P_1 = P \cup \{c\}$ be the common refinement of P and $\{c\}$, and $P_2 = P_1 \cap [a,c]$ and $P_3 = P_1 \cap [c,b]$. Then we have

$$S(P_1, f) = S(P_2, f) + S(P_3, f).$$

Since the limit of $S(P_1, f)$ exists as $\|P\| \to 0$, the limits of $S(P_2, f)$ and $S(P_3, f)$ also exist as $\|P\| \to 0$. Thus we take the limit on both sides and get

$$\int_a^b f(x)\,\mathrm{d}x = \int_a^c f(x)\,\mathrm{d}x + \int_c^b f(x)\,\mathrm{d}x.$$

□

Theorem 2.19. Let f, g be integrable on $[a,b]$ and $f(x) \geq g(x)$ for all $x \in [a,b]$. Then

$$\int_a^b f(x)\,\mathrm{d}x \geq \int_a^b g(x)\,\mathrm{d}x.$$

Proof. Note that proving this theorem is equivalent to proving

$$\int_a^b f(x) - g(x)\,\mathrm{d}x \geq 0.$$

For all tagged partitions P^*, we have

$$S(P^*, f-g) = \sum_{i=1}^{n} (f-g)(x_i^*)(x_i - x_{i-1}) \geq 0$$

Hence by the order rule, the integral is non-negative. □

Corollary 2.20.

$$-\int_a^b \left|f(x)\right|\mathrm{d}x \leq \int_a^b f(x)\,\mathrm{d}x \leq \int_a^b \left|f(x)\right|\mathrm{d}x \iff \left|\int_a^b f(x)\,\mathrm{d}x\right| \leq \int_a^b \left|f(x)\right|\mathrm{d}x$$

Theorem 2.21 (Integral Mean Value Theorem)**.** Let f be continuous on $[a,b]$. Then there exists a $c \in (a,b)$ such that

$$\frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x = f(c)$$

Proof. Since f is continuous on $[a,b]$, by the Extreme Value Theorem there are $m = \min_{x\in[a,b]} f(x)$ and $M = \max_{x\in[a,b]} f(x)$ such that

$$m \leq f(x) \leq M \text{ for all } x \in [a,b].$$

Then we have

$$m(b-a) \leq \int_a^b f(x)\,\mathrm{d}x \leq M(b-a) \implies m \leq \frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x \leq M.$$

By the Intermediate Value Theorem, there exists a $c \in [a,b]$ such that

$$\frac{1}{b-a}\int_a^b f(x)\,\mathrm{d}x = f(c).$$

□

Theorem 2.22. If f is integrable on $[a,b]$, then f is countinuous at some $c \in (a,b)$.

Proof. Assume that f is integrable on $[a,b]$. There exists a P_1 such that

$$U(P_1,f) - L(P_1,f) < 1.$$

Hence, there exists $I_1 = [x_{i-1}, x_i]$ such that

$$\sup_{[x_{i-1},x_i]} f(x) - \inf_{[x_{i-1},x_i]} f(x) < \frac{1}{b-a}.$$

Otherwise, we would have

$$U(P_1,f) - L(P_1,f) \geq \sum_{i=1}^{n} \frac{1}{b-a}(x_i - x_{i-1}) = 1,$$

By adding a point to make a refinement if necessary, we can choose the closed interval I_1 strictly contained in $[a,b]$. We have shown that f is integrable on $I_1 \subset [a,b]$. By induction there exists a partition P_n of I_{n-1} such that

$$U(P_n,f) - L(P_n,f) < \frac{|I_{n-1}|}{n(b-a)}$$

where I_n is strictly contained in I_{n-1} with

$$\sup_{I_n} f(x) - \inf_{I_n} f(x) < \frac{1}{n(b-a)}.$$

By the Nested Interval Property, there exists a $c \in \bigcap_{n=1}^{\infty} I_n$. We will show that f is continuous at c. For all $\varepsilon > 0$, take $n > 1/\varepsilon(b-a)$, then $c \in I_n$. Assume that $I_n = [p,q]$ and let $\delta = \min(c-p, q-c)$. Then whenever, $|x-c| < \delta$, we have $x \in I_n$ and hence

$$|f(x) - f(c)| \leq \sup_{I_n} f(x) - \inf_{I_n} f(x) < \frac{1}{n(b-a)} < \varepsilon.$$

□

Corollary 2.23. If f is integrable on $[a,b]$ and $f(x) > 0$, then $\int_a^b f(x)\,\mathrm{d}x > 0$.

Proof. We know that $f(x)$ is continuous at some $c \in (a,b)$. Hence for $\varepsilon = f(c)/2 > 0$, there exists $\delta > 0$ such that $x \in (c-\delta, c+\delta)$ implies

$$|f(x) - f(c)| < \varepsilon = \frac{f(c)}{2}.$$

In other words, $f(x) > f(c)/2$ for all $x \in (c-\delta, c+\delta)$. Then

$$\int_a^b f(x)\,\mathrm{d}x \geq \int_{c-\delta}^{c+\delta} f(x)\,\mathrm{d}x > \int_{c-\delta}^{c+\delta} \frac{f(c)}{2}\,\mathrm{d}x = f(c)\delta > 0.$$

□

Theorem 2.24. If f is integrable on $[a,b]$, then f^2 is integrable on $[a,b]$.

Proof. Note that $|f| \leq M$ is bounded on $[a,b]$. Pick a partition P such that $U(P,f) - L(P,f) < \varepsilon/(2M+1)$. For any $x, y \in [a,b]$, we have

$$|f^2(x) - f^2(y)| = |f(x) + f(y)| \cdot |f(x) - f(y)| \leq 2M|f(x) - f(y)|.$$

Taking the supremum over $x \in [x_{i-1}, x_i]$ and infimum over $y \in [x_{i-1}, x_i]$ gives

$$\sup_{x\in[x_{i-1},x_i]} f^2(x) - \inf_{y\in[x_{i-1},x_i]} f^2(y) \leq 2M(\sup_{x\in[x_{i-1},x_i]} f(x) - \inf_{y\in[x_{i-1},x_i]} f(y)).$$

Hence,

$$\begin{aligned} U(P,f^2) - L(P,f^2) &= \sum_{i=1}^n (\sup_{x\in[x_{i-1},x_i]} f^2(x) - \inf_{y\in[x_{i-1},x_i]} f^2(y))(x_i - x_{i-1}) \\ &\leq 2M \sum_{i=1}^n (\sup_{x\in[x_{i-1},x_i]} f(x) - \inf_{y\in[x_{i-1},x_i]} f(y))(x_i - x_{i-1}) \\ &= 2M(U(P,f) - L(P,f)) < \varepsilon. \end{aligned}$$

This shows that f^2 is integrable on $[a,b]$. □

Corollary 2.25. If f and g are integrable on $[a,b]$, then fg is integrable on $[a,b]$.

Proof. This follows from the fact that $fg = \frac{1}{2}((f+g)^2 - f^2 - g^2)$ and the linearity of integration. □

※ Numerical Integration

Definition 3.1 (Left rule)**.** We tag $x_i^* = x_{i-1}$ and denote $L_n = S(P^*, f)$.

Definition 3.2 (Right rule)**.** We tag $x_i^* = x_i$ and denote $R_n = S(P^*, f)$.

Definition 3.3 (Midpoint rule)**.** We tag $x_i^* = \frac{x_{i-1}+x_i}{2}$ and denote $M_n = S(P^*, f)$.

Theorem 3.4. Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Assume that f' exists and is bounded on $[a, b]$.

$$|f'(x)| \leq K$$

Then we have

$$\left| \int_a^b f(x) - f(a)(b-a)\,\mathrm{d}x \right| \leq \frac{K(b-a)^2}{2}$$

Proof. By the Mean Value Theorem, for each $x \in [a, b]$, there exists $c_x \in (a, x)$ such that

$$f(x) - f(a) = f'(c_x)(x-a).$$

Therefore, we have

$$\begin{aligned}
\left| \int_a^b f(x) - f(a)(b-a)\,\mathrm{d}x \right| &= \left| \int_a^b f'(c_x)(x-a)\,\mathrm{d}x \right| \\
&\leq \int_a^b |f'(c_x)||x-a|\,\mathrm{d}x \\
&\leq K \int_a^b (x-a)\,\mathrm{d}x \\
&= K\frac{(b-a)^2}{2}.
\end{aligned}$$

□

Corollary 3.5. If $[a, b]$ is partitioned into n subintervals of equal length, then we have

$$\begin{aligned}
\left| \int_a^b f(x)\,\mathrm{d}x - L_n \right| &= \left| \sum_{i=1}^n \int_{x_{i-1}}^{x_i} f(x)\,\mathrm{d}x - \sum_{i=1}^n f(x_{i-1})(x_i - x_{i-1}) \right| \\
&= \sum_{i=1}^n \left| \int_{x_{i-1}}^{x_i} f(x)\,\mathrm{d}x - f(x_{i-1})(x_i - x_{i-1}) \right|
\end{aligned}$$

$$
\begin{aligned}
&\leq \sum_{i=1}^{n} \frac{K(x_i - x_{i-1})^2}{2} \\
&= \sum_{i=1}^{n} \frac{K}{2}\left(\frac{b-a}{n}\right)^2 \\
&= \frac{K(b-a)^2}{2n}
\end{aligned}
$$

Same argument applies to R_n.

Theorem 3.6. The error bound for the midpoint rule is

$$
\left| \int_a^b f(x)\,\mathrm{d}x - (b-a) f\left(\frac{a+b}{2}\right) \right| \leq \frac{K_2(b-a)^3}{24}
$$

where K_2 is a constant such that $|f''(x)| \leq K_2$ for all $x \in [a,b]$.

Proof. By Taylor's theorem, for each $x \in [a,b]$, there exists $\xi_x \in (a,x)$ such that

$$
f(x) = f\left(\frac{a+b}{2}\right) + f'\left(\frac{a+b}{2}\right)\left(x - \frac{a+b}{2}\right) + \frac{f''(\xi_x)}{2}\left(x - \frac{a+b}{2}\right)^2.
$$

Therefore, we have

$$
\begin{aligned}
&\left| \int_a^b f(x)\,\mathrm{d}x - (b-a) f\left(\frac{a+b}{2}\right) \right| \\
=&\left| \int_a^b f'\left(\frac{a+b}{2}\right)\left(x - \frac{a+b}{2}\right) + \frac{f''(\xi_x)}{2}\left(x - \frac{a+b}{2}\right)^2 \mathrm{d}x \right| \\
=&\left| \int_a^b \frac{f''(\xi_x)}{2}\left(x - \frac{a+b}{2}\right)^2 \mathrm{d}x \right| \\
\leq& \int_a^b \frac{|f''(\xi_x)|}{2}\left(x - \frac{a+b}{2}\right)^2 \mathrm{d}x \\
\leq& \int_a^b \frac{K_2}{2}\left(x - \frac{a+b}{2}\right)^2 \mathrm{d}x \\
=&\frac{K_2(b-a)^3}{24}.
\end{aligned}
$$

□

Theorem 3.7. Suppose $f \in C^2([a,b])$ and $|f''(x)| \leq K_2$ for all $x \in [a,b]$. Then

$$
\left| \int_a^b f(x)\,\mathrm{d}x - M_n \right| \leq \frac{K_2(b-a)^3}{24n^2}
$$

Proof. Using the error bound for the midpoint rule, we have

$$\begin{aligned}
&\left|\int_a^b f(x)\,\mathrm{d}x - M_n\right| \\
=&\left|\sum_{i=1}^n \int_{x_{i-1}}^{x_i} f(x)\,\mathrm{d}x - \sum_{i=1}^n (x_i - x_{i-1}) f\left(\frac{x_{i-1}+x_i}{2}\right)\right| \\
=&\sum_{i=1}^n \left|\int_{x_{i-1}}^{x_i} f(x)\,\mathrm{d}x - (x_i - x_{i-1}) f\left(\frac{x_{i-1}+x_i}{2}\right)\right| \\
\leq&\sum_{i=1}^n \frac{K_2(x_i - x_{i-1})^3}{24} \\
=&\sum_{i=1}^n \frac{K_2}{24}\left(\frac{b-a}{n}\right)^3 \\
=&\frac{K_2(b-a)^3}{24n^2}.
\end{aligned}$$

□

Theorem 3.8. Suppose $f \in C^2([a,b])$ and $|f''(x)| \leq K_2$ for all $x \in [a,b]$. Then

$$\left|\int_a^b f(x)\,\mathrm{d}x - T_n\right| \leq \frac{K_2(b-a)^3}{12n^2}$$

Proof. We first consider the case $T_1 = (b-a)(f(a)+f(b))/2$. Define auxiliary functions

$$\begin{aligned}
g(x) &= \int_a^x f(t)\,\mathrm{d}t - (x-a)\left(\frac{f(a)+f(x)}{2}\right) \\
h(x) &= g(x) - \left(\frac{x-a}{b-a}\right)^3 g(b).
\end{aligned}$$

We have $h(a) = h(b) = 0$. By Rolle's theorem, there exists $c \in (a,b)$ such that $h'(c) = 0$. Therefore, we have

$$\begin{aligned}
0 &= h'(c) \\
&= g'(c) - \frac{3(c-a)^2}{(b-a)^3} g(b)
\end{aligned}$$

Also, we have

$$g'(x) = f(x) - \frac{f(a)+f(x)}{2} - (x-a)\frac{f'(x)}{2}$$

$$
\begin{aligned}
&= \frac{f(x)-f(a)}{2} - (x-a)\frac{f'(x)}{2} \\
&= -\frac{f''(\xi_x)}{4}(x-a)^2 \quad \text{for some } \xi_x \in (a,x).
\end{aligned}
$$

Therefore, we have

$$
\begin{aligned}
\left|\int_a^b f(x)\,\mathrm{d}x - T_1\right| &= |g(b)| \\
&= \frac{(b-a)^3}{3}|g'(c)| \\
&\le \frac{K_2(b-a)^3}{12}.
\end{aligned}
$$

The general case is derived by

$$
\begin{aligned}
&\left|\int_a^b f(x)\,\mathrm{d}x - T_n\right| \\
=&\left|\sum_{i=1}^n \int_{x_{i-1}}^{x_i} f(x)\,\mathrm{d}x - \sum_{i=1}^n (x_i - x_{i-1})\frac{f(x_{i-1})+f(x_i)}{2}\right| \\
=&\sum_{i=1}^n \left|\int_{x_{i-1}}^{x_i} f(x)\,\mathrm{d}x - (x_i - x_{i-1})\frac{f(x_{i-1})+f(x_i)}{2}\right| \\
\le&\sum_{i=1}^n \frac{K_2(x_i - x_{i-1})^3}{12} \\
=&\sum_{i=1}^n \frac{K_2}{12}\left(\frac{b-a}{n}\right)^3 \\
=&\frac{K_2(b-a)^3}{12n^2}.
\end{aligned}
$$

□

Corollary 3.9. If $f''(x) > 0$, then

$$
M_n \le \int_a^b f(x)\,\mathrm{d}x \le T_n
$$

Example 3.10 (Simpson's rule)**.** Denote $S_n = S(P^*, f)$. Then we have

$$
\begin{aligned}
S_n &= \sum_{i=1}^n (x_i - x_{i-1})\frac{f(x_{i-1}) + 4f\left(\frac{x_{i-1}+x_i}{2}\right) + f(x_i)}{6} \\
&= \frac{1}{3}(2T_n + M_k)
\end{aligned}
$$

$$= \frac{1}{3}(T_k + 2M_k)$$

where $n = 2k$.

Theorem 3.11 (Newton's method)**.** Let $f \in C^2$ and c be a root of $f(x) = 0$. Define the sequence $\{x_n\}$ by

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

If $M|x_0 - c| < 1$ where

$$M = \sup |f''(x)| \cdot \sup \frac{1}{|2f'(x)|}$$

Then $\lim_{n\to\infty} x_n = c$.

Proof. Using the Taylor expansion of f at x_n, we have

$$f(x) = f(x_n) + f'(x_n)(x - x_n) + \frac{f''(\xi_x)}{2}(x - x_n)^2$$

Since $f(c) = 0$, we have

$$\begin{aligned} 0 &= f(x_n) + f'(x_n)(c - x_n) + \frac{f''(\xi_c)}{2}(c - x_n)^2 \\ &= \frac{f(x_n)}{f'(x_n)} + (c - x_n) + \frac{f''(\xi_c)}{2f'(x_n)}(c - x_n)^2 \\ &= -x_{n+1} + c + \frac{f''(\xi_c)}{2f'(x_n)}(c - x_n)^2 \end{aligned}$$

Therefore,

$$x_{n+1} - c = \frac{f''(\xi_c)}{2f'(x_n)}(c - x_n)^2 \leq M|x_n - c|^2 \leq M^3|x_n - c|^4 \leq \cdots \leq \frac{(M|x_0 - c|)^{2^n}}{M}.$$

Finally,

$$M|x_0 - c| < 1 \implies \lim_{n\to\infty} x_n = c.$$

□

References

[1] Williams, David. *Probability with Martingales*. Cambridge University Press, 1991. Print.